Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/22/2024
PI Name:	Sanford, Larry Ph.D.		
Project Title:	Impact of Inflight Stress and Sleep Disturbances on Brain Function, Neural Communication, and Inflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	11/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per L. Juliette (Ed., 8/5/25) NOTE: End date changed to 09/30/2025 per L. Juliette (Ed., 5/8/22)		
Key Personnel Changes/Previous PI:	July 2021 report: Shao-Hui Chuang and Richard Ciavarra are no longer working on the project. July 2020 report: None. August 2024 report: None.		
COI Name (Institution):	Britten, Richard Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
Grant/Contract No.:	80NSSC19K1582		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts may experience several physical and psychological challenges including prolonged exposure to microgravity, potential social isolation (SI), and predicted exposure to ~13 cGy/Yr Space Radiation (SR). Each of these stressors can have deleterious effects on physical and mental health as well as negatively impact mission performance. Each also can disturb sleep, which can further exacerbate the effects of stress. Determining how astronauts respond to the single and compound effects of mission related stressors is crucial for understanding and mitigating factors that could impede mission success. However, examining their effects will be complex because they have different time courses and durations. SR exposure will be episodic whereas microgravity will be constant and SI and sleep disturbances can be temporary or prolonged. Their effects may also be additive or synergistic, and may vary with sex and with individual differences in stress resilience and vulnerability. Therefore, our goal in this project is to determine the single and synergistic effects of inflight stress and SR on mission related biological and functional challenges that astronauts may face. We will determine how order of experience impacts the effects of inflight stressors and determine sex and individual differences in stress resiliency using an established Wistar rat model. We will use established ground-based models of microgravity (hindlimb unloading), SI, and SR. We will use telemetry to record sleep and to assess temperature increases as an index of the stress response. We will use plasma levels of brain derived neurotropic factor to predict individual differences in stress responses. We will assess fear learning and extinction to determine how chronic inflight stressors impact ability to respond to an additional acute stressor. We will analyze food entrained, anticipatory activity to assess potential impairments in temporal perception. We will determine the effects of inflight stress and SR on sensorimotor function and recognition memory using behavioral tests. We will use local field potential recordings in the amygdala, medial prefrontal cortex, and hippocampus to determine whether stressors and SR alter the coordinated neural activity underlying fear learning and extinction, and we will determine whether there are structural changes within connections between regions. We will use chemokine/cytokine arrays and Ingenuity Pathway Analysis to determine the effect of stressors and SR on the immune system. Collectively these studies will assess functional, mechanistic, and immune consequences of several stressors inherent in space exploration. The data they provide will aid in risk assessment, and may help to develop countermeasures based on individual differences in ability to withstand stress and/or targeting immune pathways that may confer protection against the deleterious effects of multiple stressors that astronauts will face.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will increase understanding of how individual differences in resilience and vulnerability factor in determining lasting outcomes of stressful experiences. They will also determine the role that stress-induced alterations in sleep play in mediating the outcomes of stress. The results will have importance for understanding how individuals cope with stress on Earth and on missions in space.
Task Progress:	Our goal in this project is to determine the impact and potential synergistic effects that multiple inflight stressors and exposure to space radiation (SR) can have on biological systems and mission related performance. Outcomes that we are examining include sensorimotor function and balance (adhesive removal task and balance beam performance), recognition memory (novel object recognition task), time perception (anticipatory activity), fear memory, and fear memory related neural communication. We are also examining the relationship of altered performance to the effects of inflight stressors on sleep, and how responses to stressors vary in resilient (Res) animals that maintain sleep after experiencing stress and vulnerable (Vul) animals that have reduced sleep after experiencing stress. This is the fifth year of this project. Over the past year, we have continued to analyze and integrate data from our studies. We have now completed analysis and prepared manuscripts for studies examining how SR, social isolation (SI), and SR + SI impact fear memory, the stress response and sleep in male rats. These data demonstrate that SR, SI, and their combination can alter multiple aspects of stress responses and fear memory that also can vary with putative stress resilience and vulnerability. In general, SR and SR + SI increased freezing in both Res and Vul rats and differentially impacted temperature by increasing stress-induced hyperthermia (SIH) in SR Res rats to parallel higher levels observed in CTRL-Vul and SR-Vul rats. SR and SI, alone and in combination, also altered sleep amounts and architecture with differential effects in Res and Vul rats. Enhanced freezing has been considered to indicate improved fear memory and a beneficial effect of SR. However, this interpretation is problematic when enhanced freezing is considered in the context of the role fear learning and memory have in adaptive and pathological outcomes. Freezing in SR rats begins at a higher level and continues at a high level throughout shock training, thereby suggesting that their propensity to freeze is initially higher than in control rats. The continued high levels of freezing across days when footshock was no longer presented indicate significant impairment in fear extinction in SR rats. In this case, continued high levels of freezing could interfere with acquiring or exhibiting appropriate responses as the fearful/stressful situation changes. Thus, increased freezing after SR and SR + SI would likely indicate an impediment in adaptive learning, not an improvement. The impaired extinction after SR and SR + SI could also indicate a pathological change in the regulation of fear memory as the failure of fear extinction has been linked to failed coping and psychopathology. In addition to overt behaviors, an initial stressor and related fearful memories can induce similar physiological responses including increased SIH, HPA activation, and corticosteroid release, as well as increases in respiration and heart rate. SIH has a time course that parallels that of HPA activation and persists as long as psychologically stressful situations last. In this study, we found significant SIH in response to the shock training (ST), context (CTX), and extinction (EXT) phases of conditioned fear (CF) with variations across treatment groups, phenotype, and time. Compared to CTRL and SI alone, SR and SR + SI produced consistent increases in SIH in response to ST, CTX, and EXT with differences across the Res and Vul phenotypes. In general, SR promoted a Vul-like SIH response in Res rats. We determined the effects of SR on sleep in rats trained in CF and whether these varied with Res and Vul phenotypes using our established animal model of stress resilience and vulnerability. Supporting previous findings, our results showed subsequent REM responses directionally differ in Res and Vul animals following ST compared to baseline, and that phenotypes are maintained after exposure to SR. Interestingly, we also found that SR increased REM sleep and had a greater negative impact on fear memory processing, as indicated by greater freezing behavior throughout CF. By comparison, SI had greater effects on NREM sleep that were somewhat attenuated in SR + SI animals. These results indicate a broad impact of SR, SI and their interactions across multiple systems (i.e., fear memory, stress, and sleep regulation). The effects on sleep also parallel prior findings that SI beginning after SR can attenuate some SR effects on multiple behavioral and physiological systems.

	With respect to the overall project, we have completed studies on SR alone, SI alone, and SI beginning after SR in both males and females. We also have made significant progress in analyzing the data from these studies that will be needed to conduct comparisons across sexes. We expect to work on these comparisons over the coming year. The past year also has been one of accomplishment and significant challenges. We established hindlimb unloading (HLU) in our lab but were unable to conduct planned studies of sleep due to the high mortality rate in animals that received both HLU and the implant surgery needed for recording. We also began a cross-species validation study in mice and encountered a similar obstacle with respect to surgical mortality in SR-treated mice. These are different species and different treatments but do indicate that HLU and SR can negatively impact the ability of animals to withstand surgical stress. In summary, our data to date demonstrate both single and synergistic effects of SI and SR on several behaviors and sleep. They also are beginning to provide a framework for understanding how they can impact multiple, interacting systems (i.e., fear memory, stress, and sleep).
Bibliography Type:	Description: (Last Updated: 03/16/2026)
Abstracts for Journals and Proceedings	Sanford LD, Wellman LL, Luyo ZNM, Heerbrandt RS, Boden AF, Singh N, Britten RA. "Sex differences in the effects of space radiation and social isolation on sleep and EEG spectra in rats." 2024 NASA Human Research Program Investigators' Workshop. Galveston, Texas, February 13-6, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop. Galveston, Texas, February 13-6, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Luyo ZNM, Heerbrandt RS, Boden AF, Namrata S, Hwang JJ, Britten RA, Wellman LL, Sanford LD. "Behavioral sex differences in rats exposed to space radiation and social isolation." 2024 NASA Human Research Program Investigators' Workshop. Galveston, Texas, February 13-6, 2024. Absracts. 2024 NASA Human Research Program Investigators' Workshop. Galveston, Texas, February 13-6, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Sanford LD. "Mission risks and astronaut health and performance: Challenges in understanding the effects of spaceflight stress." 1st Korea-US Space Health Symposium, Incheon, South Korea, September 14-16, 2023. Abstracts. 1st Korea-US Space Health Symposium, Incheon, South Korea, September 14-16, 2023. , Sep-2023
Articles in Peer-reviewed Journals	Adkins AM, Luyo ZNM, Gibbs AJ, Boden AF, Heerbrandt RA, Gotthold JD, Britten RA, Wellman LL, Sanford LD. "Alterations in blood-brain barrier integrity and lateral ventricle differ in rats exposed to space radiation and social isolation. " Life. 2024 May 16;14(5):636. https://doi.org/10.3390/life14050636 ; PubMed PMID: 38792656; PubMed Central PMCID: PMC11122575 , May-2024
Articles in Peer-reviewed Journals	Adkins AM, Boden AF, Gotthold JD, Harris RD, Colby EM, Britten RA, Wellman LL, Sanford LD. "Effects of social isolation and galactic cosmic radiation on fine motor skills and behavioral performance." Life Sci Space Res (Amst). 2024 May;41:74-9. https://doi.org/10.1016/j.lssr.2024.01.005 ; PMID: 38670655 , May-2024